

Mode of Examination: Online
Midterm M.Sc. Semester- I Examination, 2021

Subject: Computer Science

Paper Code & Name: CSMC103 & ADVANCE COMPUTER ARCHTECTURE

Full Marks: 30

Time: 1:30 Hours

Date: 02.02.2022

Please follow the following instructions carefully.

- **Promise not to commit any academic dishonesty.**
- **Candidates are required to answer in their own words as far as applicable.**
- **Each Page of answer scripts should have your examination Roll Number on the right side top corner of your answer script**
- **The name of the scanned copy of the assessment script will be of the following format:**

• **(Example: CSMC103-MID-My Roll No.pdf)**

- **The subject of the mail should be only the assessment-script file name.**
- **The scanned assessment script is to be sent to cucse2020@gmail.com.**
- **The Assessment script will be discarded if received after the specified date and time.**
- **Write the answers with a black ink ball pen.**
- **Candidates are to check the readability of the scanned copy of the Assessment Script**

1.
 - i. What do you mean by 'instruction execution'?
 - ii. How will you justify the sizes of different memories under memory hierarchy?
 - iii. What is a Vector Processor?
 - iv. What is microcode, and why was it developed? [4x1.5=6]
2. **Answer any four questions:**
 - i. Define degradation factor and estimate it concerning branch instructions for a pipelined architecture? Clearly state your assumptions.
 - ii. In general, why the synthesized value of MAL lies in between theoretical lower and upper limits of MAL and for a given reservation table of pipelined architecture? Justify your answer.
 - iii. Split out the design steps to get a reservation table for a given latency cycle. Explain with an example, latency cycle (2,3,2,5).
 - iv. Consider an array with 100 elements, and each element occupies four words. A 32-bit word cache is used and divided into a block of 8 words. What is the hit ratio for the following statement if row major order and column major order is used?

for (i=0 ; i < 10 ; i++)
for (j=0 ; j < 10 ; j++)
A[i][j] = A[i][j] + 10;
 - v. Consider the cache has 4 blocks. For the memory references-
5, 12, 13, 17, 4, 12, 13, 17, 2, 13, 19, 13, 43, 61, 19
What is the hit ratio for the following cache replacement algorithms-
 - a. FIFO
 - b. LRU
 - c. Direct mapping
 - d. 2-way set associative mapping using LRU [4x5=20]
3.
 - i. A particular processor uses a fully associative cache of size 16 kB. The cache block size is 16 bytes. Assume that the main memory is byte-addressable and uses a 32-bit address. How many bits are required for tag and index fields, respectively, in the addresses generated by the processor?
 - ii. The size of the physical address space of a processor is 2^P bytes. The word length is 2^W bytes. The capacity of cache memory is 2^N bytes. The size of each cache block is 2^M words. For a K -way set-associative cache memory, what will be the tag field's length (in the number of bits)? [2+2]